

Christina Fan

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EDUCATION

Honours Bachelor of Science in Psychology, Neuroscience Minor 2023–2027
University of Waterloo, Waterloo, Canada

GPA: 4.0/4.0

Honours Thesis: What Makes Something Simple? Logic or Statistics? A Pattern Recognition Study
(in progress)

PUBLICATIONS

Schechter J., Gallagher D., Andreazza A. C., Black S., Cao X., Charles J., **Fan C.**, Gao F., Herrmann N., MacIntosh B., Marotta G., Marzolini S., Mills M., Mitchell S., Oh P., Rajji T. K., Rapoport M., Siddiqui M., Lanctôt K. L. “Optimizing multimodal neuromodulation in mild cognitive impairment and mild dementia: A causal mechanism for aerobic exercise priming to prevent vascular stress.” In preparation, 2026.

POSTERS

1. **Fan C.**, Wong M., Schechter J., Gallagher D., Andreazza A. C., Black S., Cao X., Charles J., Gao F., Herrmann N., MacIntosh B., Marotta G., Marzolini S., Mitchell S., Oh P., Rajji T., Rapoport M., Siddiqui M., Lanctôt K. L. “Vascular Medication Profile Moderates the Cognitive Effects of Exercise-Primed tDCS”. Sunnybrook Summer Student Poster Competition, Aug. 2026 (Toronto, ON, Canada)
2. Sun Y., Mungalsingh M., Chen X., **Fan C.**, Thompson B. “Harnessing neuroplasticity to boost 3D vision in adults with amblyopia”. Pursuits in Applied and Theoretical Science (PATH) Symposium 2026, April 2026 (Waterloo, ON, Canada)
3. Viqar U., **Fan C.**, Perrier M., Graham K., Vaart J., Meehan S., Barnett-Cowan M. “Cortical mechanisms underlying audiovisual integration in healthy aging: A TMS study”. Lake Ontario Visionary Establishment (LOVE) Conference, Feb. 2026 (Niagara Falls, ON, Canada)
4. **Fan, C.**, Gravestock, J., Cisorio, D. “Impact of Word Tangibility on the Effectiveness of Different Mnemonics”. Social Science Advanced Research Methods Seminar Presentation, University of Waterloo, Dec. 2025 (Waterloo, ON, Canada)

RESEARCH EXPERIENCE

Sunnybrook Health Sciences Center/ Toronto, ON May 2026–Aug. 2026

Summer Student, Geriatric Neuropsychopharmacology Research Group

Principal Investigator: Dr. Krista Lanctôt

- EXPRESS-V Project: Examining effects of exercise-primed tDCS on cognition of patients at risk of MCI/vascular dementia
- Patient recruitment, literature review, administered and scored cognitive assessments, statistical analysis

- Developed experience with R, SPSS, Covidence, and clinical neuropsychological batteries

University of Waterloo/ Waterloo, ON

Research Assistant, Human Visual Neuroscience Lab

Sep. 2025–Present

Principal Investigator: Dr. Ben Thompson

- VELUX Study: Exploring effects of tDCS on reading skills in patients with visual impairments
- Administer tDCS on patients as per study protocols, perform data preprocessing

Research Student, Face Processing and Social Cognition Lab

Jan. 2026–Apr. 2026

Principal Investigator: Dr. Roxane Itier

- Ran independent weekly EEG sessions examining neural and behavioral responses to emotional facial expressions and attentional processes
- Data collection using EEG, eye-tracking (Eyelink), and running MATLAB scripts
- Stimuli creation using GNU Image Manipulation Program (GIMP), EEG data preprocessing

Research Student, Multisensory Brain and Cognition Lab

Sep. 2025–Apr. 2026

Principal Investigator: Dr. Michael Barnett-Cowan

- Multisensory Integration Project: Administering TMS on older adults to explore changes in audiovisual integration during aging
- Participant recruitment, ran TMS sessions, and performed statistical analyses using Excel

Research Assistant, Psychological Interventions Research Team

May 2025–Dec. 2025

Principal Investigator: Dr. Jonathan Oakman

- Coded 400+ clinical therapy session segmentations using standard guidelines
- Attended weekly meetings to improve therapeutic ratings

Research Assistant, Cognition and Natural Behavior Lab

Oct. 2024–Dec. 2025

Principal Investigator: Dr. Evan Risko

- Scheduled and ran 40+ study sessions independently using SONA
- Performed data coding for five large datasets

University of California, Berkeley/ Remote

May 2025–Jul. 2025

Research Assistant, Berkeley Early Learning Lab

Principal Investigator: Dr. Fei Xu

- Study material preparation, participant recruitment, and testing
- Coded 100+ infant eye-gaze data videos using LookIt and Datavyu software

TECHNICAL SKILLS

Programming languages and tools: Python, R, MATLAB, Bash, Git

Data analysis programs: SPSS, jamovi, Excel

Experiment programming: PsychoPy, Qualtrics

Neuroimaging Techniques and Apparatus: transcranial magnetic current stimulation (tDCS), transcranial magnetic stimulation (TMS), Electroencephalography (EEG), functional near-infrared spectroscopy (fNIRS), EyeLink (eye-tracking system)

VOLUNTEER EXPERIENCE

Volunteer Jan. 2026–Present
Waterloo Regional Health Network, Hospital Elder Life Program

- Weekly shifts delivering standardized cognitive interventions to patients experiencing delirium
- Monitor and chart patient symptoms and interactions to update individual care plans
- Collaborate with healthcare staff to provide dietary, physical, and cognitive support to patients while adhering to hospital standards

Literacy Tutor Jul. 2019–Present
Toronto Public Library, Leading to Reading Program

- Tutor children in grades 1-6 with literacy difficulty
- Plan weekly reading comprehension, writing exercises, and literacy games
- Monitor and record child progress, assign homework

Retail Director Jun. 2024–Jul. 2025
University of Waterloo, Psychology Society

- Managed inventory and finances of retail and merchandise
- Trained and scheduled volunteers for shifts on campus Tuck Shop
- Maintained order and cleanliness of Tuck Shop

HONOURS AND AWARDS

University of Waterloo, Faculty of Science 2023–Present
Term Distinction

Service Ontario Nov. 2023
Ontario Volunteer Service Award

University of Waterloo Sep. 2023
President's Scholarship